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Challenges in biodegradation of non-degradable thermoplastic waste: from environmental impact to operational readiness

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Abstract

Non-degradable plastics such as polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET) are among the most generated plastic wastes in municipal and industrial waste streams. The mismanagement of abandoned plastics and toxic plastic additives have threatened marine and land fauna as well as human beings for several decades. The available thermal processes can degrade plastic at pilot- and commercial-scale. However, they are energy-intensive and can generate toxic gases. Degradation of plastic waste with the help of live microorganisms (biodegradation) is an eco- and environmentally friendly method for plastic degradation, although the slow processing time and low degradation rate still hinder its applications at pilot- and large-scale. In this review, the advantages and limitations of current plastic degradation methods, their technology readiness levels (TRL), biodegradation mechanisms and the associated challenges in biodegradation are assessed in detail. Based on this analysis, a path toward an efficient and greener way toward degradation of non-recyclable single-use PE, PP, PS and PET plastic is proposed.

Keywords

Biodegradation; plastic waste; pre-treatment; biosurfactant, environmental impact; technology readiness levels

1. Introduction

Plastics are long-chain polymers that have strong water resistance, strength, durability and lightness, which made them ideal alternatives to cellulosic-based materials (Muhonja et al. 2018). Due to the superior properties of plastics, they are one of the most popular materials used in consumer and industrial products. The broad usage of plastics ranges from pharmaceuticals, food packaging, cosmetics, to beverage industries. The world production of plastic as of 2018 is about 360 million tonnes per year, of which about 3 % is dumped directly into the ocean, and from the remaining, 79 % is landfilled, 12 % is thermally incinerated and just 9 % is recycled (Ritchie and Roser 2018).

Plastics are categorised as thermoplastics and thermoset. Thermoplastics are a type of plastic that can be deformed upon heating and reform again once they are cooled. This characteristic of thermoplastic makes them be flexible and recyclable several times, nevertheless, the quality will reduce eventually (Cabanes Gil et al. 2019). Thermoset plastics, on the contrary, are stronger and brittle than thermoplastics but they cannot be reformed after the initial heating (Sharma et al. 2019). Despite that thermoset plastics are widely used in automotive, lighting and electrical industries, the scope of this review is more focused on the treatment and degradation of waste thermoplastics.

The most globally generated thermoplastic wastes which also are the main topic of this review are polyethylene terephthalate (PET), high density polyethylene (HDPE), low density polyethylene (LDPE), polypropylene (PP) and polystyrene (PS) with the proportion of about 11, 14, 20, 19 and 6 % in municipal waste stream, respectively (Fig 1) (Shah et al. 2008, Parker 2018, Fotopoulou and Karapanagioti 2019).

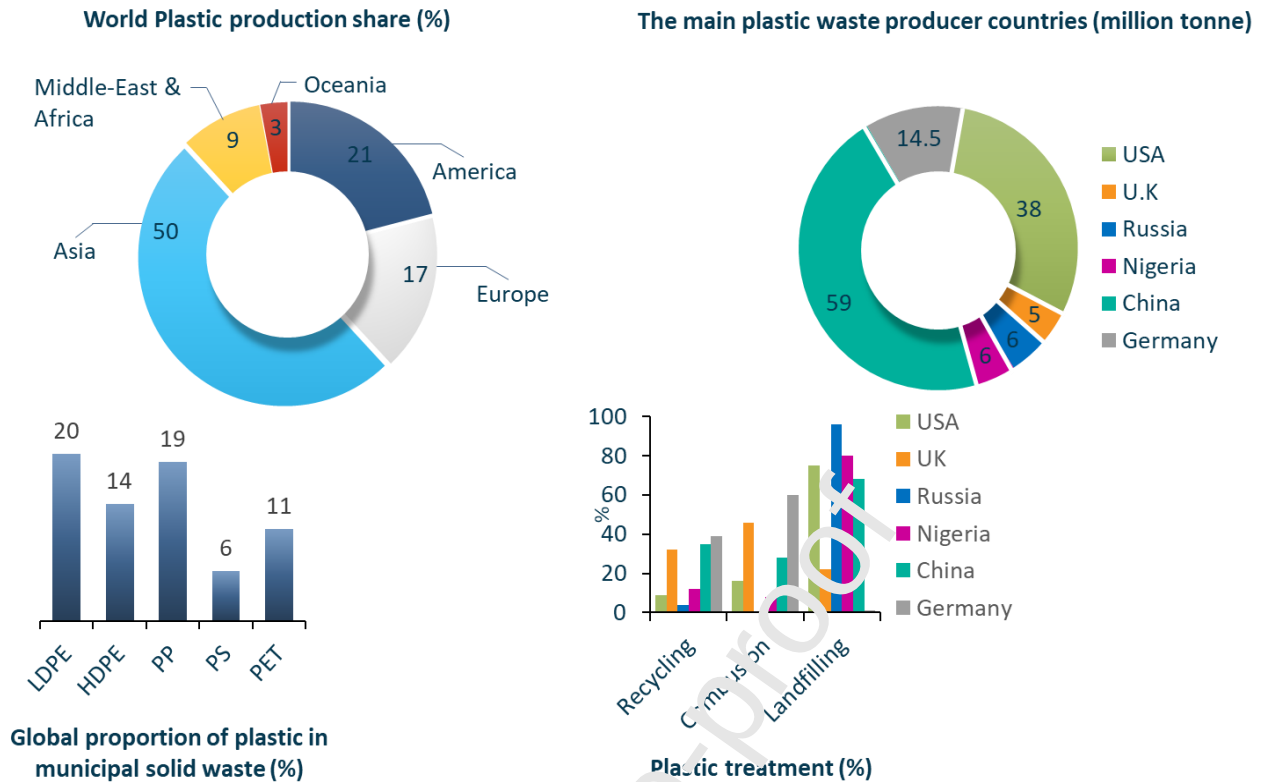


Fig 1. The estimated share of global plastic waste and the used treatment method in main plastic waste producer countries (Ritchie and Rosen 2018, Royer et al. 2018, Fedotkina et al. 2019, Olanrewaju and Oyebade 2019, Jiang et al. 2020)

Depending on the type of plastic and the environmental conditions, it is estimated that 10-1000 years is required for the natural degradation of thermoplastic wastes. The careless abandonment of plastic waste in the environment and deficiency of a proper waste treatment system would cause serious problems on land and marine animals. such as trapping, suffocation, and ingestion (Thevenon et al. 2014, Jambeck et al. 2015).

Plastic wastes are generated from post-consumer and post-industrial applications (Ignatyev et al. 2014). The post-industrial wastes are those plastic wastes that are disposed of at the end of a production line, and will never reach consumers. Such plastic wastes are easier to manage and handle due to their known types, sizes, and clean nature. On the other hand, post-consumer plastic wastes are mainly generated after a single use of a plastic product or plastic-containing product. It is noteworthy that post-consumer plastic wastes are usually mixed type and contaminated by other

municipal solid waste streams; therefore, their separation, recycling and degradation require more energy and time (Hubo et al. 2016).

No matter where plastic waste stems from, its accumulation in the environment can cause serious issues. The presence of plastic debris in the aquatic system would release toxic compounds like plasticiser, bisphenol, phthalates, ethers, etc. into the soil, water system, and food chain (Meeker et al. 2009). These compounds can be harmful to the land and marine wildlife and endanger human health. Every year about 400,000 to one million people are killed in developing countries due to the burning of mismanaged municipal waste containing plastic debris and its associated diseases. It is also estimated that over 100 million just marine animals are killed every year from plastic pollution (Calderwood 2019, Harvey 2019, Ismail 2019). The production and treatment of plastic waste contribute to global warming because 3.8 % of the gasses produced from the processes are greenhouse gasses (methane, ethylene, and carbon dioxide) (Royer et al. 2018, Zheng and Suh 2019). It is estimated that increasing the population and quadrupling demand for plastic by 2050 will result in the emission of 56 billion tonnes of greenhouse gases (GHG), which is 13 to 15 % of the entire carbon budget (Lisa et al. 2012).

To degrade and deconstruct plastic wastes, biological and non-biological treatment methods can be used. The non-biological treatment includes photo-oxidation (ex. ultraviolet), thermal-oxidation (ex. incineration, pyrolysis, and gasification), hydrothermal liquefaction, and chemical hydrolysis (ex. enzymes, acid, base, and solvent). Mechanical processes while are not considered as a degradation method, are usually applied before each treatment to enhance the deconstruction efficiency. Recycling of plastic usually involves chemical or thermal processes that infuses with mechanical forces for the conversion of plastic waste into new products. Recycling is most feasible when the type of plastic is known, and plastic wastes all be in one type. Moreover, for some types of thermoplastics (e.g., PP and PS) which are coated with anti-oxidants, UV resistant, flame-retardant, and stabiliser to have a long-life usage in different applications (such as vehicle parts and electronic

appliances) recycling can be the best eco and environmentally friendly method. However, recycling of single-use plastics which are mostly mixed type and often contaminated with different municipal organic waste (e.g., food debris) may not be the proper method due to the high labour and energy requirement (e.g., collection, separation, and washing steps).

Thermal and chemical processes while effective in the degradation of plastic waste at a commercial scale within a short period, often have other limitations and drawbacks. These methods are labour and energy-intensive, release perilous chemicals and toxic gases into the environment, require a high volume of chemicals, and have difficulties in the recovery of precious by-products (Ragaert et al. 2017, Moharir and Kumar 2019). Despite technological advances and serious efforts in plastic degradation, a gap in utilisation of a standard and environmentally friendly method of plastic degradation remains. On the contrary, biodegradation will not leave hazardous gases or chemicals in the environment and is not energy-intensive in comparison to the current conventional methods (Schink et al. 1992, Urbanek et al. 2018).

The scope of this review is to provide an overview of the advantages and limitations of current treatment methods for degradation of PE, PP, PS, and PET plastics, evaluate the capabilities of microorganisms and mechanisms in biodegradation, and assess the environmental impact of plastic degradation. Technology-readiness level was used to further understand whether biodegradation of plastic on a commercial scale is feasible or not.

2.1. Landfilling

It has been estimated that around 79% of plastic wastes are sent to landfill sites (Ritchie and Roser 2018). Generally, landfilling can be conducted in open or closed (sanitary) format. In open-type landfilling, post-consumer plastic debris which is usually mixed with other organic compounds is accumulated on land without any engineering design of dumping site. The abandonment of plastic waste in the open-type landfill would go under the natural photo-oxidation process. The UV and temperature from the Sun make plastic to be more brittle by defragmenting some of the C-H bonds within the polymeric structure (Andrady 2011, Raquez 2011). Consequently, the polymeric building blocks, additives and plasticizers are released into the environment over time (Asakura et al. 2004, Hammer et al. 2012, Iskander Syeed Md 2016). Moreover, the emitted greenhouse gases (such as CH₄, CO₂, N₂, N₂O, SO₂ and H₂S) from this type of landfilling have direct impacts on global warming and ozone depletion (Lamb D 2012, Royer et al. 2018). Eriksson and Finnveden (2009) reported 253 g CO₂ / kg of disposed plastic is emitted in the open-type landfill in Northern Europe. In a study by Royer et al. (2018) the amount of released CH₄ to the atmosphere from both virgin and aged non-degradable plastics were calculated, in which the amount of discharge CH₄ to the atmosphere was between 10- 4100 mol/g day.

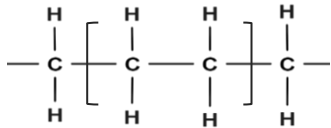
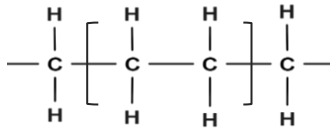
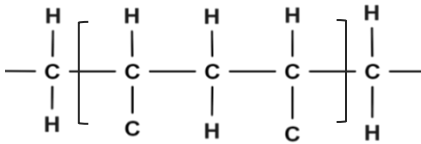
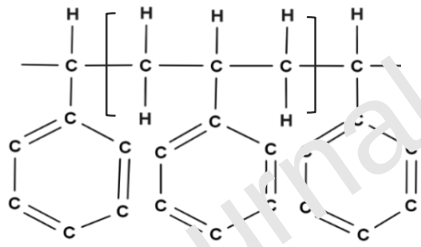
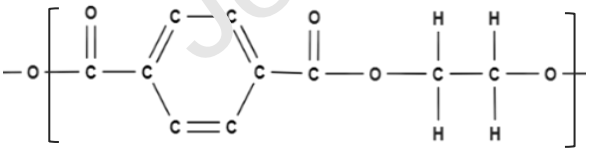
In closed landfilling known as sanitary plastic and other municipal solid waste are buried into a pit with thick protective layers to minimise leaching into soil and water resources (Zafirakou A et al. 2013, Patil et al. 2017). Due to the lack of light and oxygen, degradation via photo and thermal oxidation is not feasible (Carballa et al. 2007). Although the lack of oxygen in sanitary landfilling, activate anaerobic strains, biodegradation of non-degradable plastic via this way is almost zero due to the very low energy production compared to aerobic condition. Iwańczuk et al. (2015) showed that the degradation rate as weight loss and mechanical properties of the buried PE in anaerobic conditions were not changed within 72 days of incubation. Even though landfilling compare to

Other treatment methods is the easiest and most common way of disposal, it is restricted in many countries due to the the environmental and land regulations.

2.2. Incineration

Incineration is a thermal process in which air and high temperatures ($>800^{\circ}\text{C}$) are used to reduced more than 90 % of plastic waste (Demirbaş 2000, Al-Salem et al. 2009). Incineration is also known as a waste-to-energy method that is used globally to reduce about three quarter million metric tonnes of solid waste per day to generate heat energy (Makarichi et al. 2018). A study by Bujak (2015) showed that an engineered incineration unit can significantly recover the produced heat energy from the combustion of plastic waste. Energy recovery of incineration is dependent on the calorific value of the waste. Different properties of solid waste (paper, food, plastic etc.) will provide different calorific values and energy. The higher the calorific value, the higher the energy recovered from its combustion. Among other solid waste, oil-based plastic can provide higher calorific value upon combustion, which makes them a good source for energy recovery (Table 1) (Al-Salem et al. 2009). Morris et al., (1996) reported that effective and complete incineration of plastic waste can generate almost three times the energy compared to combustion of other municipal solid waste. Scott (1999) stated that mixed plastics have a calorific value between 30-40 MJ/kg which is almost three times higher than other municipal solid waste with calorific value of 10 MJ/kg.

Table 1. Calorific value of plastic from combustion, emitted toxic gases, and GHG yield as CO₂e (Wasielewski and Siudyga 2013, Zheng and Suh 2019)

Plastic type	Structure	Calorific value (MJ/kg)	CO ₂ e emission* (kg CO ₂ e/tonne polymer)	Toxic gases from combustion
LDPE		40- 44	3050	Dioxins
HDPE		37-40	3071	Furans
PP		41-44	3349	Carbon monoxide
PS		38-40	4757	Carbon dioxide
PET		21-23	4137	Benzene
Mixed-plastic (Σ PE, PS, PP, PET)	-	35-40	3673	Formaldehyde
				Acrolein
				Polycyclic aromatic hydrocarbons

* Over the life cycle of plastic from production to its conversion.

Incomplete combustion of plastic and other municipal solid waste can cause serious health problems due to the formation of toxic gases such as dioxins ($C_4H_8O_2$), furans (C_4H_4O), carbon monoxide (CO), carbon dioxide (CO_2), hydrogen sulphide (H_2S), heavy metals and polycyclic aromatic hydrocarbons (PAHs) (Ecke H 2008, Valavanidis et al. 2008, Verma et al. 2016). Li et al. (2001) showed that from the incineration of PE alone 462.3 mg/kg PAHs is produced which is enough to cause cancer and carcinogenic problems in humans.

Although, incineration is seemed to be an effective way of waste reduction and energy recovery, neutralisation, and reduction of generated toxic compounds is currently one of the main challenges in incineration treatment plants due to the involved costs (Su et al. 2015, Mikus et al. 2016).

2.3. Pyrolysis

Like incineration, pyrolysis is also categorised as a thermal treatment process which often is known as thermolysis (Anuar Sharuddin et al. 2016). In pyrolysis complex materials are cleaved into smaller molecules in a non-oxidative condition under elevated temperatures (500-800 °C) at above atmosphere pressure (Miandad et al. 2016). Pyrolysis has been practiced since the 19th century for main conversion of biodegradable organic compounds like wood and coal (Sharma 2019). However, the mass use of plastic and its generated waste in the environment, has caused pyrolysis to be used as a method of plastic waste degradation (Anuar Sharuddin et al. 2016).

Liquid yield and its conversion to bio-oil/ fuel are the main aims of plastic waste pyrolysis (Scheirs and Kaminsky 2006). Polyolefin (PE and PP) and polystyrene (PS) are good candidates for pyrolysis due to their higher calorific value and high yield of liquid oil. Production of liquid oil from pyrolysis of these plastics is well studied (Demirbas 2004, Kaminsky et al. 2004, Kim and Kim 2004, Wong et al. 2017). In most cases, more than 80 wt % liquid oil can be obtained from their pyrolysis (Williams and Slaney 2007). Ahmed et al. (2014) obtained about 81 wt % liquid from pyrolysis of PE at 350 °C. Marcilla et al. (2009) also achieved about 85 wt % liquid oil from PE pyrolysis at 550°C (Marcilla et al. 2009). In other studies, 81 and 88 wt % liquid oil was

produced by pyrolysing PS and PP at 450 and 500 °C, respectively (Mhandad et al. 2010, Thamm et al. 2019).

In opposite to the high yield of liquid oil (>80 wt %) produced from PE, PP, and PS pyrolysis, generation of liquid oil from PET pyrolysis is much lower (23-40 wt %). The lower liquid-oil production of PET is due to the different composition and lower calorific value of PET than other plastics. Since PET is composed of heteroatoms (ethylene glycol and terephthalic acid) for liquid oil production more heating spectrum and operating time are required to breakdown each part of the molecule. In a study conducted by Sogancioglu (2017) the amount of liquid oil from PET pyrolysis was almost zero due to the oligomer structure of the produced oil. The low thermal conductivity of plastic, high operational cost, and energy consumption limit pyrolysis as a green and eco-friendly method of degradation.

2.4. Gasification

Gasification also is a thermal treatment method to convert solid waste such as plastic into valuable syngas (H_2 , CO). Depending on the use of oxidising agent, plastic sort, gasifier type, and operational condition, the composition and application of the produced syngas are varied (Heidenreich and Foscolo 2015). The process occurs in presence of oxidising agents (air, steam, or O_2) at elevated temperatures (500-1300 °C) (Young 2010, Dai et al. 2015). One of the main advantages of gasification compared to incineration and pyrolysis is its flexibility in using mixed feedstock (ex. mixture of plastic with other solid waste). Moreover, the amount of required oxidising agent is lower than complete ignition as such less toxic gases (ex. dioxin and furan) are emitted to the environment (Thakare and Nandi 2016, Lopez et al. 2018).

Gasification of plastic waste containing PE, PP, PS, and PET results in almost complete conversion of plastic to volatile compounds (Sharuddin et al. 2016). However, in most cases, the municipal plastic wastes are mixed with other feedstock (ex. biomass, food residue, cardboard) which generate tar and char at the end of the process. One of the major challenges in gasification is the generation of char and tar, which reduce the overall syngas production (Wilk and Hofbauer 2013). To enhance

the efficiencies of the process, an additional step (heterogeneous gasification) is often considered for the conversion of the formed char to syngas (Lopez et al. 2018).

The composition of produced syngas as mentioned earlier depends on operating conditions and the input feedstock. Gasification of plastic waste is currently conducted on the industrial scale, however, the complex behaviour of plastic scission during the operation, the tar and char formation, and high operating temperature ($< 500^{\circ}\text{C}$) are still the main obstacles of this treatment technology.

2.5. *Hydrothermal liquefaction*

Hydrothermal liquefaction (HTL) is a subdivision of hydrothermal treatment occurring in a non-oxidative condition that has been used for depolymerisation and conversion of waste to oil (Behrendt F 2008, Toor 2011). The process occurs in presence of water which plays a key role in breaking down complex compounds at subcritical (pressure: 15 to 100 bar, temperature: 150 to 250 $^{\circ}\text{C}$) or supercritical (pressure > 221 bar, temperature > 374 $^{\circ}\text{C}$) conditions (Brunner 2009, Toor 2011, Baroutian et al. 2013, Hii et al. 2014, Baroutian et al. 2016).

HTL is relatively less energy-intensive in comparison to pyrolysis, gasification, and incineration since a pre-treatment step like drying is not required (Schindler 2015, Reißmann et al. 2018). Yang et al. (2017) studied the energy requirement between pyrolysis and HTL. As a result, they found that HTL required about 1.6 times lower energy consumption compared to pyrolysis. HTL also generates a lower tar and char compared to pyrolysis and incineration, which makes it a good substitute to conventional thermal waste degradation methods (Peterson AA 2008, Gollakota et al. 2018, Kumar 2018). HTL is mainly used for biomass compounds, however, several studies were conducted so far on the conversion of plastic waste to oil by HTL technology (Shah et al. 1999, Zhang et al. 2007, Sugano et al. 2009, Wang et al. 2014, Raikova et al. 2019). The quantity and quality of produced oil in HTL is highly important and depends on several factors like the input feedstock, operating temperature, processing time, and utilisation of homogenous or heterogeneous catalysts (Akhtar 2011). In the study conducted by Mansur et al. (2018) different parameters were examined for higher fuel production from plastic waste. Chen et al. (2019), obtained 91 wt % oil

from liquefaction of PP plastic waste under a supercritical condition, which was about 10 wt % higher than oil yield in pyrolysis.

Despite the lower energy requirement of HTL for breaking down the polymeric structure and lower char/ tar formation compared to pyrolysis and incineration, the high operating conditions still are one of the important capital and operational limitations.

3. Biological plastic degradation

Biodegradation of plastics in this context defines as utilisation of live microorganisms (mainly bacteria/ fungi) for depolymerisation of plastics and their consumption as a food source. Compared to thermal and hydrothermal degradation methods, the production of valuable products like bio-oil and syngas currently is not feasible in this method due to the low speed of biodegradation, and the nature of involved processes. However, biodegradation is relatively cheaper than the methods like pyrolysis and incineration due to not operating at high operating conditions. Moreover, biodegradation does not leave toxic gases and hazardous compounds in the environment which makes it as an interesting eco and environmentally friendly way of plastic degradation.

The first sign of microbial colonisation on plastic dates back to the 1970s (Carpenter and Smith 1972). Since then, the search for identification of plastic degrading microorganisms has never stopped, and several strains from the genus of *Rhodococcus*, *Aspergillus*, *Pseudomonas*, *Penicillium*, *Bacillus*, *Serratia*, *Phanerochaete*, *Rhizopus* were reported to have the capabilities of degrading PE, PP, PS, and PET (Iiyoshi et al. 1998, Sivan et al. 2006, Kyaw et al. 2012, Yang et al. 2014, Yoshida et al. 2016, Lou et al. 2020, Zhang et al. 2020, Taghavi et al. 2021). As it can be seen from Fig 2, the number biodegradation studies were increased from 33 in 1970-1980 to about 670 in 2010-2020. The studies were mainly focused on PE plastic which possibly could be due to the inert structure of PE that hardly degradation in the environment, and due to its higher production demand compared to other plastic types.

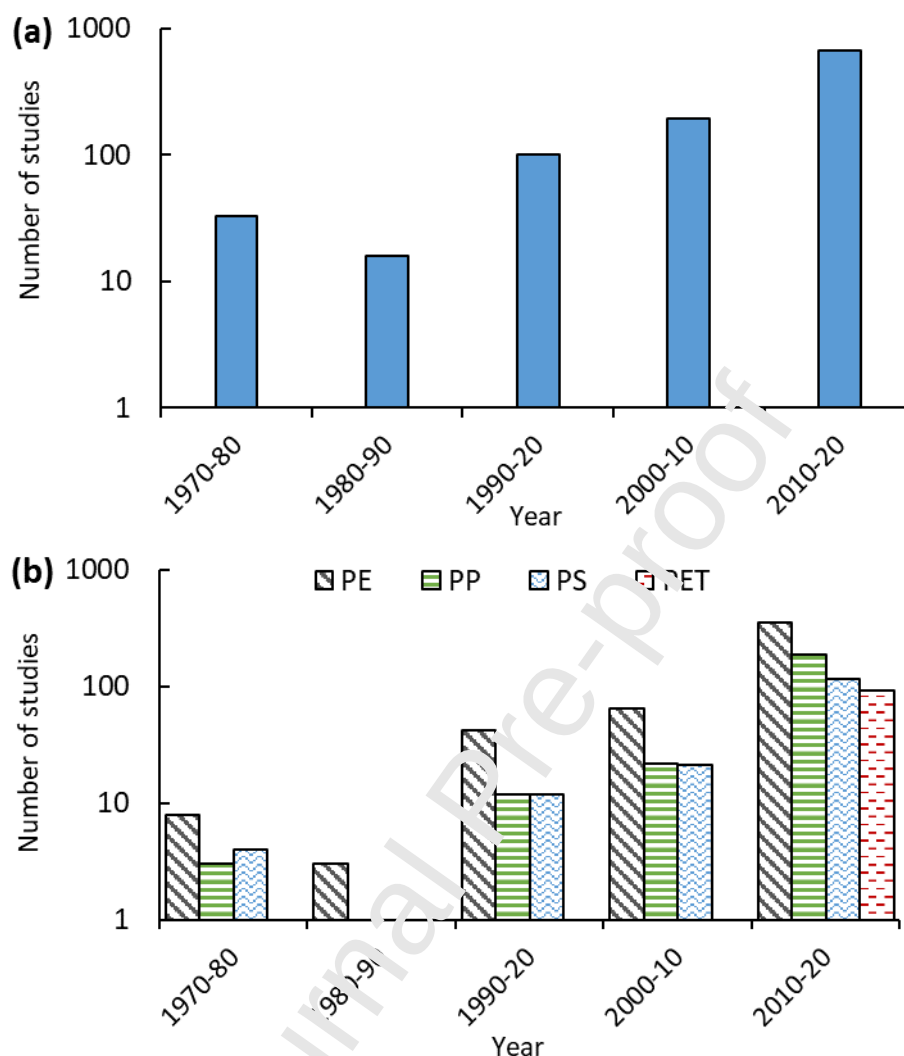


Fig 2. The number of biodegradation studies from 1970-2020 for (a) all non-degradable plastic types; (b) each plastic type. The data was obtained from Scopus database by limiting keywords in the abstract as “Biodegradation of plastic” AND “plastic type”.

The increase in the number of studies in the last decade was due to the significant outcome, and identification of microbial capabilities in colonisation and secretion of enzymes to degrade plastic. For example, *Bacillus* spp. was reported to play an important role in plastic biodegradation due to their ability to secrete surface active-compounds (biosurfactant) that results in higher plastic

accession numbers and microbial colonisations (Mukherjee et al. 2010, Vimala and Madhew 2010, Tadimeti 2020). Recently, Yoshida et al. (2016), identified a novel strain *Ideonella sakaiensis* 201-f6 that able to degrade PET plastic into its building blocks via secretion of PETase and MHETase. Identification of the aforementioned new enzymes caused other researchers specifically in recent years to focus more on PET biodegradation via enzyme engineering (Wei et al. 2016, Knott et al. 2020). In Table 2 the capability of some of the highlighted strains in biodegradation of each plastic type is illustrated in more detail.

Table 2. The capabilities of some highlighted strains of plastic biodegradation

Strains	Plastic type	Degradation time (day)	Weight loss (%)	Reference
<i>Aspergillus niger</i>	PE	30	17.35	(Kathiresan 2003)
<i>Aspergillus niger</i>	PE	30	4.32	(Wadood et al., 2018)
<i>Aspergillus niger</i>	PE	160	-	(Raghavendra et al. 2016)
<i>Aspergillus niger</i>	PE	30	5.8	(Raaman et al. 2012)
<i>Aspergillus versicolor</i>	PE	90	40.6	(Gajendiran et al. 2017)
<i>Aspergillus flavus</i>	PE	28	3.9	(Zhang et al. 2020)
<i>Aspergillus flavus</i>	PE	100	5.5	(Taghavi et al., 2021)
<i>Aspergillus japonica</i>	PE	30	11.11	(Raaman et al., 2012)
<i>Aspergillus fumigatus</i>	PE	30	2.49	(Wadood et al. 2018)
<i>Aspergillus glaucus</i>	PE	30	28.8	(Kathiresan, 2003)
<i>Bacillus brevis borstelensis</i> 707	PE	30	12	(Hadad et al. 2005)
<i>Bacillus amyloliquefaciens</i>	PE	60	11	(Das and Kumar 2015)
<i>Bacillus subtilis</i> UCP 999	PET	60	9.06	(Jara et al. 2009)
<i>Bacillus subtilis</i>	PE	30	9.26	(Vimala and Mathew 2016)
<i>Bacillus subtilis</i>	PE	60	17.7-23.1	(Ibiene et al. 2013)
<i>Bacillus subtilis</i>	PS	30	23	(Mohan et al. 2016)
<i>Bacillus subtilis</i>	PE	30	1.75	(Harshvardhan and Jha 2013)
<i>Bacillus pumilus</i>	PE	30	1.5	(Harshvardhan & Jha, 2013)
<i>Bacillus mixture</i>	PE	30	-	(Abrusci et al. 2011)
<i>Bacillus cereus</i>	PE	40	1.6	(Roager and Sonnenschein 2019)
<i>Bacillus cereus</i>	PP	40	12	(Auta et al., 2017)
<i>Bacillus cereus</i>	PS	40	7.4	(Roager and Sonnenschein 2019)
<i>Bacillus cereus</i>	PET	40	6.6	(Roager & Sonnenschein, 2019)
<i>Bacillus gottheilii</i>	PS	40	5.8	(Roager and Sonnenschein 2019)
<i>Bacillus gottheilii</i>	PE	40	6.2	(Roager and Sonnenschein 2019)
<i>Bacillus gottheilii</i>	PET	40	3	(Roager & Sonnenschein, 2019)
<i>Bacillus circulans</i>	PE	45	2.33	(Aly et al. 2018)
<i>Bacillus mycoides</i>	PE	60	10.5-11.3	(Ibiene et al., 2013)
<i>Bacillus sphaericus</i>	PE	365	3.5-10	(Sudhakar et al. 2008)
<i>Bacillus thuringiensis</i>	PE	9	-	(Ray 2019)
<i>Bacillus</i> sp. YP1	PE	28	10.7	(Yang et al. 2014)
<i>Ideonella sakaicnsis</i> 201-f6	PET	42	58	(Yoshida et al., 2016)
<i>Penicillium simplicissimum</i>	PE	90	38	(Sowmya et al. 2015)
	(treated)			
<i>Penicillium pinophilum</i>	PE	942	-	(Volke-Sepúlveda et al. 2002)
<i>Penicillium frequentans</i>	PE	60	0.45	(Seneviratne et al. 2006)
<i>Penicillium funiculosum</i>	PET	84	0.21	(Nowak et al. 2011)
<i>Penicillium chrysogenum</i> NS10	PE	90	34.3- 58.5	(Ojha et al. 2017)
	(treated)			
<i>Penicillium oxalicum</i> NS4	PE	90	36.6 – 55.3	(Ojha et al., 2017)
	(treated)			
<i>Pseudomonas aeruginosa</i>	PE	120	20	(Kyaw et al. 2012)
<i>Pseudomonas aeruginosa</i>	PE	30	7.3- 8.5	(Badahit et al., 2018)
<i>Pseudomonas syringae</i>	PE	120	11.3	(Kyaw et al., 2012)
<i>Pseudomonas putida</i>	PE	120	9	(Kyaw et al., 2012)

<i>Pseudomonas citronellensis</i>	PE	4	17.8	(Shadd et al. 2014)
<i>Pseudomonas fluorescens</i>	PE	30	7.8 -7.9	(Badahit et al. 2018)
<i>Pseudomonas</i> sp.	PE	30	20.54	(Kathiresan, 2003)
<i>Rhodococcus rubber</i> C208	PS	56	0.86	(Mor and Sivan 2008)
<i>Rhodococcus rubber</i> C208	PE (treated)	56	7.5	(Sivan et al. 2006)
<i>Rhodococcus rhodochrous</i>	PE	240	-	(Eyheraguibel et al. 2017)
<i>Sporosarcina globispora</i>	PP	40	11	(Auta et al. 2017)

3.1. Mechanism of biodegradation

The mechanism of biodegradation ties to the microbial tactic to turn insoluble and long-chain molecules into soluble and small molecules through deterioration, depolymerisation, mineralisation, and assimilation steps.

The biodegradation begins with the adhesion and colonisation of living microorganisms (bacteria/fungi) to the surface of plastics (deterioration). The surface deterioration then would ease the penetration and colonisation of microorganisms internally. In the second stage, extracellular microbial exoenzymes further break down the deteriorated plastic into various compounds with different molecular weights (Mueller 2003).

In the assimilation step, some of the small medium size oligomers (less than 600 Daltons) formed during the previous steps are now able to be transported or diffuse into cytoplasm of microbes to be consumed for anabolism or catabolism. However, the remaining dimers and oligomers which could not be utilised formerly, are required to be further broken down by microorganism's catalytic agents (exoenzymes / free radicals) (Lucas et al. 2008). Now, some of the formed molecules are small and soluble enough to pass through the semi-permeable microbial membrane and be used as their energy sources. Upon complete degradation, the final products of mineralisation step based on microbial respiration are CO₂, H₂O, and CH₄ (Gu 2003, Kolvenbach et al. 2014). Nevertheless, other metabolic products such as microbial biomass, salt, and organic compounds can also be formed as a result of biodegradation (Dommergues and Mangenot 1972, Muhonja et al. 2018). In Fig 3, the schematic of the plastic biodegradation mechanism is illustrated.

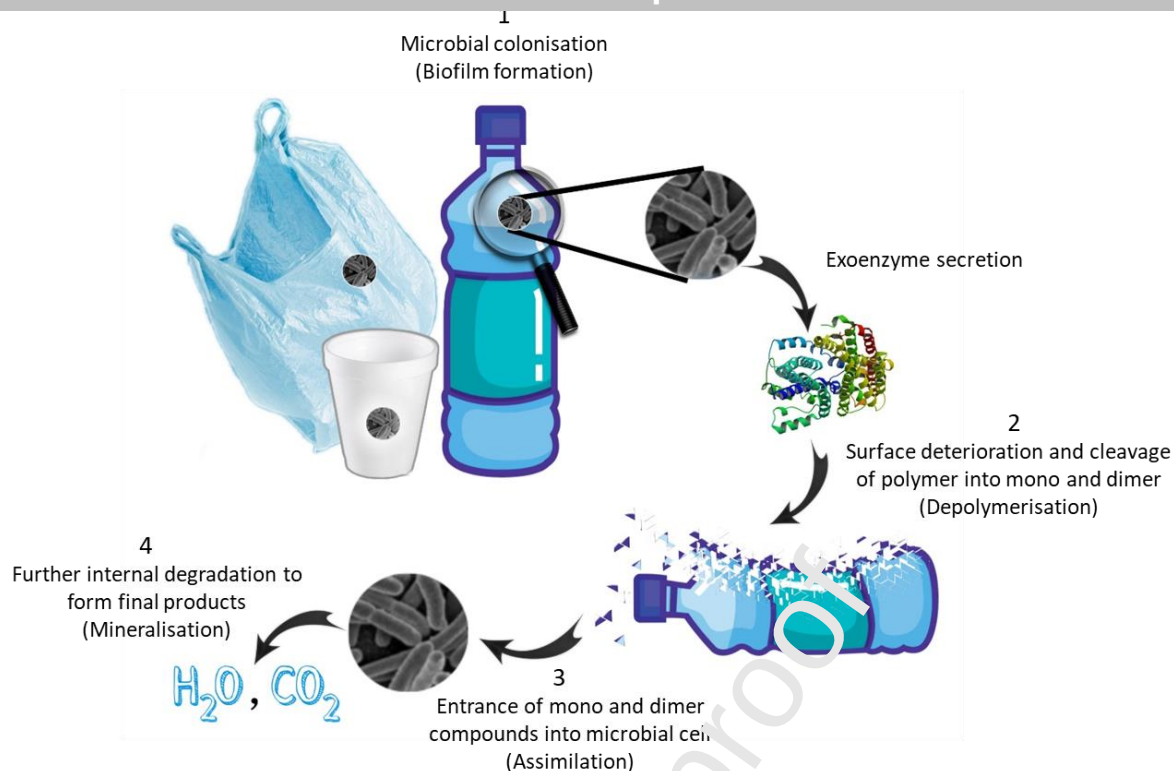


Fig 3. The mechanism of plastic degradation by microorganisms

As mentioned before, biodegradation is highly linked to the secretion of effective microbial exoenzymes on the surface of plastic (Donatuma 1991, Kamal M R and Huang B 1992). Indeed, microbial enzymes act as a catalytic agent to break down long-chain and high-weighted polymers into smaller fragments. Their role is to reduce the activation energy for speeding the reactions (Blanco and Blanco 2017). Sometimes, for breaking down compounds, secretion of an enzyme from one strain is not effective and efficient enough, and instead, synthesis of several enzymes from a group of strains (microbial consortia) is needed.

It has been reported that the degradation activity of selected strains as a mixed community is faster and more efficient than the degradation of monoculture due to the participation of each strain in the microbial community and the diversity of secreted enzymes (Sowmya et al. 2015, Skariyachan et al. 2019). The utilisation of microbial consortia rather than a single strain for biodegradation enhancement was also reported by several researchers (Manzur et al. 2004, Roy et al. 2008, Mukherjee et al. 2016). The biodiversity of strains derived from landfill, soil, sludge, or wastewater can significantly affect the biodegradation process. In the study conducted by Taghavi et al. (2021)

higher surface erosion and physical alteration were observed on PS and PET plastic where these plastics were incubated with a mixture of strains (*Penicillium raperi*, *Aspergillus flavus*, *Penicillium glaucoroseum*, and *Pseudomonas* spp.) rather than a single strain. Ogunbayo et al. (2019) showed that the incubated PE with a mixture of *Pseudomonas* sp. and *Aspergillus niger* degraded up to 15 wt% compared to the individual incubation of PE with *Pseudomonas* spp (7.2 wt%) and *Aspergillus niger* (12.4 wt%).

Secretion of microbial enzymes is either constitutive or inductive. In the former case, enzymes exist constantly in the microbial pathways, while in the latter case the catalytic enzymes are involved in the reactions only if an inducing signal or external initiator reaches the cells. The synthesis of inductive enzymes is increased over time until the inducer is used completely (Lucas et al. 2008). A common example of an inductive enzyme is the synthesis of β -galactosidase for hydrolysing of lactose into D-glucose and D-galactose in *Escherichia coli*. The production of β -galactosidase enzyme in *E. coli* was induced due to the presence of lactose in the microbial media as the sole carbon source. Indeed, the *E.coli* cells cannot utilise the lactose to synthesize β -galactosidase when glucose is available insufficiently in the media (Juers et al. 2012). While degradation of a complex polymer such as plastic is different and not comparable to degradation of simple sugar like lactose because stimulation can sometimes induce signals in microbial cells pathways to secrete peroxidase enzymes in benefit of polymer biodegradation, such as PE plastic. Peroxidase is produced in most aerobic organisms to oxidise H_2O_2 to water and oxygen during aerobic respiration. It has been reported that stimulation by a non-lethal concentration of H_2O_2 can induce the microbial cells to secrete peroxidase for H_2O_2 removal. Moreover, Iiyoshi et al. (1998) showed that for breaking down the lignin in the media, fungal cells are required to secrete lignin peroxidase, which then was in benefit of PE biodegradation.

While the exact mechanism and role of peroxidase on PE degradation are not yet well understood, it is believed that the oxidation of H_2O_2 by peroxidase can cause the formation of free $OH\cdot$ and $O\cdot$ radicals. Since these radicals are not stable, they can interact with other organic compounds. In the presence of PE these radical species bind to the cleaved end or middle carbon of PE and make it

more susceptible to microbial enzymatic attack. The free OH- and O- radicals can respectively make OH-C-C-C-C- and O=C-C-C-C- upon cleavage of PE from its end or can make -C-C-OH and =O-C-C-C when PE is cleaved from its middle part.

Despite the inductive or consecutive type of enzyme, the production of microbial enzymes can be extra or intracellularly. Intracellular enzymes (endoenzymes) associate with inner microbial pathways (tricarboxylic acid cycle, beta-oxidation, etc.) while extracellular enzymes (exoenzymes) secrete outside the microbial cell; i.e., within the aquatic media and/or among the organic matters (soil, sludge, sediment) for the initial breakdown of the polymeric structure. The exoenzymes are responsible for cutting the polymeric chains from the ends and edges (Moussard 2006, Mateo et al. 2007). Usually, the excreted exoenzymes react with molecules of 10-50 carbons; however, Yoon et al. (2012) reported a possible microbial enzymatic activity on molecules having 2000 carbons.

Enzymatic depolymerisation is followed by the catalytic triad mechanism sharing a set of three coordinated amino acids (aspartate, histidine, and serine). The aforementioned catalytic triad mechanism can be found in the α/β -hydrolase fold superfamily (Stryer L et al. 2002, Sun et al. 2014). An example of the serine-histidine-aspartate catalytic triad can be found in the study conducted by Yoshida et al. (2016). In their study, a novel enzyme, PETase, was identified from *Ideonella sakaiensis* 201-F6 which was able to degrade PET plastic into mono-2-hydroxyethyl terephthalate (MHET) and a low amount of bis (2-hydroxyethyl) terephthalate (BHET). Similar to the other previous PET degrading hydrolase enzymes (lipase, esterase, and cutinase), PETase also belongs to of α/β hydrolase superfamily that uses triad serine-histidine-aspartate catalytic mechanism for depolymerisation (Liu et al. 2018).

The mechanism starts with the serine where its active site becomes receptive for the upcoming substrate in its cleft (Hedstrom 2002). The serine then makes a nucleophilic attack on the substrate and discharges its hydrogen ion, which is later taken by the facilitated histidine (step 1). This nucleophilic attack would cause that serine to behave as an alkoxide group for making a covalent bond with the substrate carbonyl carbon (step 1) (Lucas et al. 2008, Alves et al. 2018). It is worth

Remembering that the transition state complex is not formed without the cooperation of histidine and aspartate in the mechanism. Aspartate reduces the acidity level of imidazole nitrogen in histidine structure by increasing the pKa from 7 to 12; this makes histidine a strong base to take the hydrogen ion released from the nucleophilic attack of serine to the substrate (Gutteridge and Thornton 2005).

In the intermediate state, tetrahedral intermediates are produced which are then transformed into acyl-enzyme intermediate. This will be performed by a proton donation of histidine to the amide group of the substrate and in its cleavage at C-terminuses from the enzyme complex (steps 2 & 3) (Neitzel 2010). The next step in ternary catalytic hydrolases enzymes is in a correlation of water molecule. The hydroxide ion (OH^-) of the water binds to the ester functional group causing the removal of the covalent bond between serine and peptide (step 4). Subsequently, serine is turned into a very nucleophilic alkoxide group and takes the hydrogen (H^+) ion of water from histidine (step 5) (Jiang et al. 2016). This brings separation of another peptide from the enzymatic mechanism and re-generation of serine peptidase for another set of the enzymatic cycle (step 6). In Fig 4, the catalytic mechanism of triad serine, histidine-aspartate upon reaching a compound having a carbonyl group in its structure was illustrated.

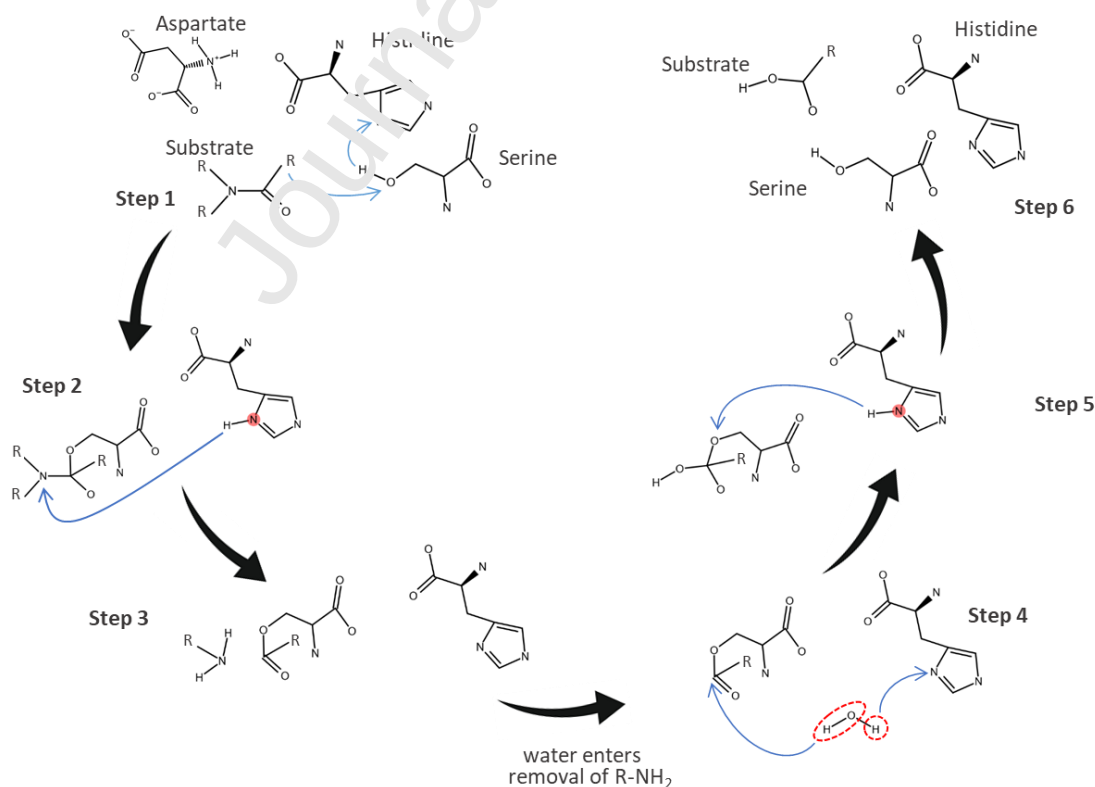


Fig 4. The catalytic mechanism of triad serine-histidine-aspartate; adapted from Cozma (2014)

3.2. Parameters influencing biodegradation

As mentioned earlier, biodegradation highly depends on the diversity of strains in the consortium, (landfill, soil, ocean, wastewater), the size of the microbial community, growth conditions (aerobic or anaerobic), colonisation on/in plastic, and secretion of different enzymes through the aforementioned mechanisms. For example, in aerobic conditions, microorganisms intake plastic as a carbon source, and use oxygen molecules in their electron transport chain to reach the final product of CO_2 , H_2O (Krzysztof and Magdalena 2012). This is how in anoxic conditions, nitrate (NO_3^-), ferric iron (Fe^{3+}), sulphate (SO_4^{2-}), and carbon dioxide (CO_2) are used as the terminal electron acceptors in the electron transfer chain to produce CH_4 , CO_2 , or H_2O as end-products. Compared to anaerobic respiration, more energy known as adenosine triphosphate (ATP) is produced in aerobic respiration, which indeed provides higher microbial survival, reproduction, and consequently a better biodegradation efficiency.

The size of the microbial community (inoculum) is also another important factor in the biodegradation process. The higher number of microorganisms increases the chance of survival, colonisation, enzyme secretion, which can result in higher biodegradation. While no study measured the direct effect of inoculum size in biodegradation of the aforementioned thermoplastics (PE, PS, and PET), the effect of different inoculation size was studied in bioremediation of other tyre and toxic compounds like phenol and bisphenol-A (the usual toxic compounds derived from degradation of plastic). Bessey et al. (2002) showed that an increase in the size of *Acinetobacter* sp. from 0.05 g/L to 0.2 g/L could increase biodegradation of phenol by 8 folds. Similarly, Lao et al. (2019) also found that the maximum specific growth rate and shortest lag phases of *Bacillus* sp. on scrap tyre occurred when inoculum size increase from 5 to 20 %. In another study, Eltoukhy et al. (2020) found that the inoculum size lower than the optimum value reduced microbial survival, colonisation, and enzymes synthesis for degradation of bisphenol-A.

Besides the role of microorganisms in the biodegradation process, physiochemical properties of plastics (e.g., size, shape, thickness, and crystallinity) also can play crucial roles in benefit or

against biodegradation. Chamas et al., (2020) described in detail that a plastic film could be degraded 260 and 1100 times faster than plastic fibre and bead of the same mass and crystallinity respectively. Increasing the number of branches in polymers structure results in higher molecular weight, toughness, viscosity, strength, and melting point, which indeed reduces the biodegradation efficiencies. Since microorganisms' transport systems are only able to take up low-mid range molecular weight into their intracellular membrane, high-molecular-weight polymers are required to be defragmented into the smaller molecules before passing through the cell membrane (Shah et al. 2008, Kolvenbach et al. 2014).

Environmental factors such as UV, heat, weather conditions, and physical forces can cause surface cracking and erosion, chemical-bond cleavage, molecular weight, and tensile strength reduction. These physiochemical alterations make plastic more fragile and susceptible to be degraded by microorganisms. In the study conducted by Roy et al., (2008), the UV pre-treated PE degraded higher than un-irradiated PE. Velrajan & Andreu (2012) also showed that applying heat as a way of PE physicochemical alteration could enhance the weight loss up to 5 %.

4. Biodegradation challenges

One of the main challenges in biodegradation is increasing degradation efficiency. Currently, the biodegradation rates of plastics are much lower than the conventional thermal treatment technologies. The low efficiency of plastic biodegradation stems from the lack of proper microbial enzymes, and the strong physiochemical structure of plastics making the biodegradation harder (Sánchez 2020).

On top of the low degradation efficiency, the effect of plastic degradation by-products on the environment is another concern. While biodegradation is considered the most environmentally friendly way of plastic waste reduction in comparison to thermal and landfilling technologies (Shah et al. 2008), the incomplete biodegradation process can also result in the formation of various by-products that are harmful to the environment. Therefore the toxicity of created by-products is needed to be carefully assessed for any new biodegradation processes (Shahnawaz et al. 2019). In

this section, some of the common solutions for biodegradation enhancement as well as the environmental impact of produced by-products are discussed.

4.1. Pre-treatment

In pre-treatment technologies (known as plastic aging), plastic is exposed to chemical (acid, base, solvent), heat, and/or ultraviolet (UV) radiation to enhance microbial colonisation and biodegradation efficiencies (Farzi et al. 2019). Regardless of the type of initiator in pre-treatment technologies, the principal almost follows the same and would lead to the formation of free radicals. The generated radicals will participate in different steps of the polymeric degradation pathways (initiation, propagation, and termination). Depending on the plastic type, random chain scission, autoxidation, and cross-linking reactions occur to create different short compounds (Gewert et al. 2015).

In the first step, UV or heat will cause the formation of free radicals within the plastic backbone (step 1). It should be noted that for plastics such as PE and PP which lack unsaturated double-bonds in their structure, autoxidation may not occur as easily as PS and PET plastics. PE and PP are mainly produced in a transparent shape which means they lack chromophore for UV absorption. However, the addition of additives, dyes, pro-oxidant, impurities, or manufacturing defect is just enough to make PE and PP susceptible to autoxidation (Gewert et al. 2015, Chamas et al. 2020).

In the propagation step, the formed polymer radicals can join together via crosslinking or undergo autoxidation by the available environmental oxygen (step 2). An abstraction of H from another polymer can bind to the formed peroxy group in another polymer and make hydroperoxide. Chain scissions occur and the hydroperoxide converts into alkoxy radical and peroxy radical by participation in different reactions (step 3). The propagation step repeats several times and water usually is generated as a by-product. In the final step, termination, two radicals join together to form short- chain and more stable compounds. Since alkoxy radical is less stable than peroxy radical, it undergoes random chain session and results in the formation of carbonyl and alkene (Vasile and Pascu 2005).

Photo-oxidation or thermal oxidation of PS (having styrene ring in its structure) as a way of pre-treatment can lead to the formation of olefin, carbonyl, styrene, α -methyl styrene, benzene, benzoic acid, toluene, benzyl alcohol, phenol, benzaldehyde (Hoff et al. 1982). Oxidation of PS will cause a colour change from whitish to yellowish and makes PS be more brittle (Yousif and Haddad 2013). Despite PE, PP, and PS having C-C in their backbone, PET is made of heteroatoms by a combination of ethylene glycol and terephthalic acid. The existence of hydrolytic groups within the PET structure would lead PET to be hydrolysed from its ester or amide bond (Müller et al. 2001). Chain scission and hydrolysis of PET through the polymeric degradation pathway will create compounds mainly vinyl and carboxylic acid end-group (Thomas and P. M 2011). In Fig 5. the reaction mechanisms of lead to form shorter compounds are illustrated.

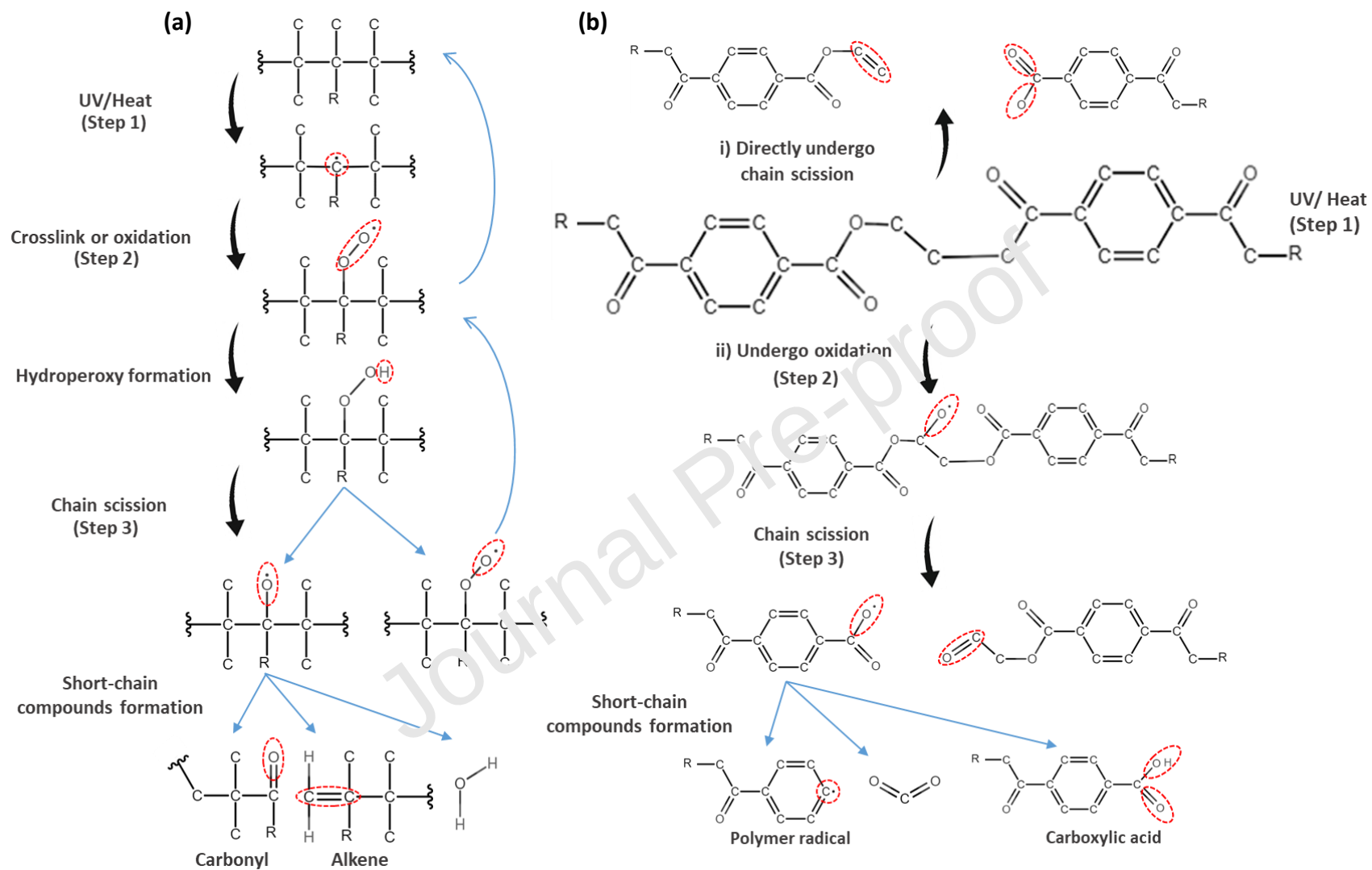


Fig 5. Degradation mechanisms of (a) PE, PP, and PS with carbon backbone structure and (b) PET with ethylene glycol and terephthalic acid;

adopted

from

Gewert

et

al.

(2015)

The creation of these short-chain compounds (carbonyl, carboxylic acid, and olefin) upon or plastic pre-treatment will increase hydrophilicity, microbial colonisation, and biodegradation efficiency (Erdmann et al. 2020). While the effect of pre-treated PS and PET were not studied in biodegradation enhancement, there were many reports on the role of pre-treated PE and PP in biodegradation improvement. Esmaeili et al. (2013), pre-treated PE plastics under UV light in a laminar hood for 25 days, and observed higher biodegradation (as mineralisation) in the UV pre-treated PE plastic (29.5 %) than un-treated plastic (15.8 %). In another study conducted by Markandan et al. (2019), higher bacterial colonisation and growth rates were observed on the UV irradiated PET than un-irradiated.

Chaudhary and Vijayakumar (2020) also reported that the chemically pre-treated PE with nitric acid for 6 days can enhance the degradation efficiency up to 71.6 %. Rajandas et al. (2012) achieved 50.5 and 61% biodegradation after giving the pre-treated PE in nitric acid to *Pseudomonas aeruginosa* and *Microbacterium paraoxydans*, respectively. In another study, Kittur and co-workers (2013) obtained weight loss up 27.33 % when pre-treated PE with UV and nitric acid was given to microorganisms. Chemical pre-treatment of plastic alone or in combination with other pretreatment methods while is effective in hydrolysing of plastic and degradation efficiency, it is not environmentally friendly. Indeed the used chemical compounds (acid, base or organic solvent) is required to be recovered totally before their discharge into the environment, which is problematic. In Table 3, some of the used pre-treatment methods for improving the biodegradation efficiency by particular strain is shown in details.

Plastic	Strain	Pre-treatment method	Pre-treatment time (day)	Degradation time (day)	Weight loss %	Reference
PE	<i>Bacillus subtilis</i>	UV (< 300 nm)	3	30	9.26	(Vimala & Mathew, 2016)
PE	<i>Bacillus</i> spp. (<i>halodenitrificans</i> , <i>pumilus</i> , <i>cereus</i>)	UV (280-370 nm)	-	730	8.4	(Roy et al. 2008)
PE	<i>Penicillium simplicissimum</i>	UV and acid treated	-	90	38	(Sowmya et al. 2015)
PE	<i>Penicillium simplicissimum</i>	UV	20.8	90	-	(Yamada-Onodera et al. 2001)
PE	<i>Penicillium simplicissimum</i>	Nitric acid	6		-	(Velrajan and Andrew 2012)
	<i>Penicillium</i> sp.	UV (< 300 nm)	10	120	~ 3.5	
	<i>Rhizopus arrhizus</i>				~ 3	
	<i>Penicillium</i> sp.				~ 5	
	<i>Rhizopus arrhizus</i>	Heat (100 °C)	30		~ 4.3	
	<i>Penicillium</i> sp.				~ 4.5	
	<i>Rhizopus arrhizus</i>	Nitric acid (concentrated)	10		~ 4	
PE	<i>Acinetobacter pittii</i>	UV (245 nm)	16	28	26.8	(Montazer et al. 2018)
PE	<i>Curvularia lunata</i> SG1	UV (354 nm)	2.1	90	43.76	(Raut et al. 2015)
		Heat (80 °C)	5		36.20	
		UV (354 nm) + Heat (80 °C)	2.1 + 5		48.40	
PE	<i>Arthrobacter oxydans</i> + <i>Arthrobacter globiformis</i>	UV (312 nm) + Pro-oxidant (calcium stearate)	15 + 7	30	52	(Pereira et al. 2012)
PE	<i>Aspergillus niger</i> & <i>Lysinibacillus splanchniticus</i> XDB9	UV (254 nm)	25	126	29.5	(Esmaeili et al., 2013)
PE	<i>Brevibaccillus borstelensis</i> strain 707	UV (312 nm)	2.5	30	11	(Hadad .D et al., 2004)
PE	<i>Fusarium</i> sp. AF4	UV (254 nm) + Nitric acid (99%)	16.4	90	-	(Hasan et al. 2007)
PE	<i>Microbacterium paraoxydans</i>	Nitric acid	10	60	61	(Rajandas et al. 2012)
	<i>Pseudomonas aeruginosa</i>				50.5	
PP	<i>Bacillus flexus</i>	Heat (100 °C)	8	365	0.65	(Arkatkar et al. 2010)
	<i>Bacillus subtilis</i>				0.6	
	<i>Bacillus flexus</i>	UV (225 nm)	6		2.5	
	<i>Bacillus subtilis</i>				1.5	

PP	<i>Bacillus plexus</i> + <i>Pseudomonas azotoformans</i>	UV (225 nm)	6	365	1.95	(Aravintnan et al., 2016)			
Table	3.	Previous	studies	on	plastic	pre-treatment	and	biodegradation	enhancement

One of the main struggles in the enhancement of biodegradation efficiency is linked to the increase of microbial colonisation on the surface of the plastic. Previous studies showed that the utilisation of mineral-oil and surfactant were able to improve the biodegradation efficiencies. Surfactants are active compounds with an amphiphilic structure that can reduce the surface tension between the plastic and microbial cells. This causes plastics to be dispersed more in the media and results in a higher microbial attachment. Mor & Siven (2008) also reported that the utilisation of mineral oil as a surface-active agent increased the microbial biofilm formation on PS plastic. In the study conducted by Gong et al. (2019), higher microbial colonisation on PE plastic incubated with Tween 80 was observed.

Unlike synthetic surfactants which are non-degradable and toxic for the environment, biosurfactants (surfactants extracted from microorganisms) are completely biodegradable and can be used in bio-based applications. Several genera of bacterial species (ex. *Bacillus*, *Pseudomonas*, *Acinetobacter*, *Rhodococcus*, *Mycobacterium*, *Serratia*, *Corynebacterium*, *Flavobacterium*, *Clostridium*) were stated as biosurfactant producers for industrial-scale applications (Bodour et al. 2003, Sachdev and Cameotra 2013). However, the utilisation of synthetic and biosurfactants in plastic biodegradation has rarely been reported, which could stem from the complexity and effectiveness of the process. Almost all of the previously conducted studies were focused just on the enhancement of PE biodegradation and underestimated the role of surface-active compounds for a higher degradation of other plastic like PP, PS, and PET.

Among the previously mentioned strains, only the biosurfactants from *Bacillus* and *Pseudomonas* spp are known as surfactin and rhamnolipid respectively were studied in details. It was reported they are efficient in biodegradation enhancement by reducing the water surface tension from 73 mN/m to less than 30 mN/m (Soberón-Chávez and Maier 2011). In the study conducted by Marajan et al. (2018) the surface tension of water reduced significantly from 73 to 30 after just 18 h growth of *Bacillus subtilis* in media.

Vinhalia and Madhew (2016) also showed that the secretion of surfactin from *B. subtilis* could increase the weight loss up to 9.2 %. From the reported results, *B. subtilis* has a great potential to be examined further in the enhancement of plastic biodegradation. Despite the potentiality of biosurfactants from *B. subtilis* in plastic biodegradation, there is still an extensive gap in the effect of many other surface-active compounds for maximising microbial attachment and biodegradation improvement of PP, PS, and PET plastic. Table 4, shows the studied surface-active compounds in plastic biodegradation.

Table 4. The utilisation of surfactant and biosurfactant for biodegradation enhancement

Plastic	Strain	Surface-active agent	Degradation time (day)	Weight loss (%)	Reference
PE	<i>Bacillus licheniformis</i> and <i>Lysinibacillus</i> sp.	Biosurfactant	30 days	2.97	(Mukherjee et al. 2016)
PE	<i>Bacillus subtilis</i>	Biosurfactant	30	9.26	(Vimala and Mathew 2016)
PE	<i>Bacillus licheniformis</i> and <i>Pseudomonas fluorescens</i>	Biosurfactant	30	7.13	(Mukherjee et al. 2018)
PE	<i>Streptomyces coelicoflavus</i> NBRC 15399T	Biosurfactant	28	30	(Midhun et al. 2015)
PE	<i>Pseudomonas aeruginosa</i>	Tween 80	60	-	(Albertsson et al. 1993)
PE	<i>Cupriavidus necator</i> H16	Tween 80	21	33.7	(Montazer et al. 2018)
PE	<i>Lysinibacillus fusiformis</i>	SDS (Sodium dodecyl sulphate)	30	7	(Mukherjee et al. 2017)
PE	<i>Rhodococcus ruber</i> C208	Mineral-oil	30	8	(Gilan et al. 2004)

In last recent decades, the research for the production of biodegradable plastic by proportional addition of degradable compounds like starch or corn into the structure of non-degradable thermoplastic was in high demand. Several studies stated that biodegradable plastics could be a green alternative to petroleum-based plastics, and degraded at a higher rate in the environment by the microbial community (Lefebvre 2012, Khoramejadian 2013, Mierzwa-Hersztek et al. 2019). Though the reported degradation efficiency of biodegradable plastics containing additives often misinterprets, and the stated degradation rates are related to the degradable part (starch or corn) rather than the plastic itself. In the study conducted by Dave et al. (1997) the blended PE with starch (70:30) showed only 6.3 % PE weight loss and 84.5 % degradation of starch. The addition of starch or corn into the plastic structure makes it more hydrophilic and results in higher microbial colonisation. Due to the gained microbial enzymes through the evolution, the biodegradable parts will degrade initially into simple sugars for consumption as a food source. However, after almost complete consumption of biodegradable parts, the degradation rate will be reduced due to the non-degradability of plastic itself. Rapa et al. (2010) examined the biodegradability of PE-starch blended (70:30) with soil microorganisms for three months. Their result showed that biodegradability is correlated to the consumption of starch and not the plastic itself. They also stated that the only benefit of starch blended PE was improvement of the penetration of microbial into plastic through the formed holes in plastic after utilisation of biodegradable parts.

Besides the aforementioned reasons, the production of biodegradable plastics would not solve the current global plastic crisis. One of the current environmental issues, is the mass abandonment of plastics and micro and nanoplastics in soil, ocean, and water systems which can endanger several marine and land creatures. To solve the present plastic problem in the world, a green and efficient method is required to degrade the current plastic waste rather than making biodegradable plastic for future use.

One of the reasons that the biodegradation treatment of plastics is still in its infancy is due to the lack of proper microbial enzymes for complete degradation. Nevertheless, to date different enzymes (ex. esterase, catalase, hydrolase, lipase, tannase, cutinase, laccase, peroxidase, laccase, PETase, and MHETase) were discovered to degrade PE, PP, PS, and PET at low rate. Among the aforementioned secreted enzymes, one of the highlighted examples is the discovery of PETase and MHETase from a novel bacterium *I. sakaiensis*. While these two enzymes are among the most powerful natural secreted enzymes for the deconstruction of PET into its building blocks, its low thermal stability at elevated temperature reduces its utilisation at any commercial scale.

To maximise the stability and degradation efficiency, enzyme engineering can be beneficial for overcoming the associated challenges. In the study conducted by Austin and co-workers (2018), an engineered mutated PETase with a wider active site was synthesised which allows binding of larger compounds. Compared to the natural PETase, their mutant enzyme was able to improve the degradation activity of both PET and its bio-derived replacement (polyethylene-2,5-furandicarboxylate).

Wei et al. (2016) also showed that, by exchanging the cutinase enzyme sequence extracted from *Thermobifida fusca* KW3 (TfCut1) to the leaf-branch compost cutinase (LLC) from compost metagenome, their engineered enzyme was able to degrade PET more than 42wt% after 50 h of hydrolysis at 65°C. In a recent study conducted by Son et al. (2019) the engineered PETase not only exhibited higher stability by 8.8 °C, but also the degradation activity improved 14-fold relative to the wild type.

Knott and co-workers (2020) recently, engineered a super enzyme by linking PETase and MHETase (PETase-MHETase) together via a bridge, which could degrade PET into its building block up to six times faster than the naturally occurring enzyme when their secreted alone.

Despite significant improvement in the production of engineered PETase for PET degradation, there is not yet effort for synthesising the mutant enzyme for PE, PP, and PS degradation. While PE, PP, and PS are also among the most generated plastic wastes globally, the unavailability of proper natural enzymes to work with could be one of the main limitations in protein engineering. One of the reasons that biodegradation's efficiency of plastic in the environment is very low can stem from the fact that in the environment (e.g., landfill) plastics are not the only source of carbon for microorganisms, and they are usually mixed with other biodegradable organic wastes. Consequently, the chance of microbial habitation on plastic as the only available food source and synthesis of a new enzyme for its degradation will reduce. To identify a new plastic degrading enzyme from naturally occurring microbes, plastics should be introduced as the sole carbon source to microorganisms for a long period. The identification of two new plastic degrading enzymes (PETase, MHETase) from the novel strain, *I. sakaiensis* was approve of the aforementioned statement. In which *I. sakaiensis* inoculated from the yard of a contaminated bottle-recycling factory which PET was the only source of carbon for five years.

4.5. Environmental impacts of generated by-products

Depolymerisation of plastic in the environment by heat, UV, or physical forces remains different un-degradable products that could affect the environment, animals, and humans. The nature of these generated by-products depends on different factors such as plastic type, strain specious, and microbial enzymatic system. In the study conducted by Shahnawaz et al. (2019) the diversity of generated by-products from two different bacteria were assessed. They found that even at the same operating condition each strain behaves differently on PE degradation, which results in the creation of different by-products. While the variety in the formation of by-products can broaden the chance of toxicity, most of the generated by-products were reported to have no or low-level toxicity to humans, animals, and the environment. Farzi et al. (2019) identified 20 different compounds from incubation of PET with *Streptomyces* sp; they found that except ethylbenzene and o-xylene the other identified by-products were safe for both humans and the environment.

In another study conducted by Sangale et al. (2015), the toxicity of derived by-products from biodegradation of PE with *Aspergillus terreus* strain MANF1/WL and *Aspergillus sydowii* strain PNPF15/TS were tested for the viability of a tiger shark and germination of sorghum seeds. In that study different compounds as 2, naphthalene carboxylic acid; 1,2 benzene dicarboxylic acid; octadecanoic acid; diethyl phthalate; dibutyl phthalate; 1,2-bis (trimethylsilyl) benzene; hexasiloxane; 2-cyclohexen-1; 7-Methylenebicyclo [3.2.0] hept-3-en-2-one; cyclohexane-1 and dodecahydropyrido [1,2-b] isoquinolin-6-one were formed with no death and germination inhibition. Pramila and Ramesh (2015) also reported that the formed 2-buthene; 2- methyl, acetone, and ethene from biodegradation of PE with *Acinetobacter baumannii* were not toxic on germination, chlorophyll content, and physical morphology of *vigna radiata*. In another study, analysis of formed by-product from incubation of *Brevibacillus* spp. and *Aneurinibacillus* spp. with PE and PP plastic within 140 days confirmed the formation of cis-2-chlorovinyl acetate; tri-decanoic acid, and octa-decanoic acid with no toxicity effect in the environment (Skariyachan et al. 2018).

Taghavi et al. (2021) used different bacteria and fungi strains to evaluate the effect of derived by-product from incubation mixed plastic (PE, PS, and PET) with soil, activated sludge and farm sludge microbial consortia. The toxicity of the derived by-products was evaluated colorimetrically by consumption of triphenyl tetrazolium chloride (TTC) in live and active strains. They found that the produced by-product had no inhibition in microbial growth and activity compared to control. In table 5, some of the highlighted generate compounds from the biodegradation of plastics by different microbial strains are shown.

Table 5. The list of produced by-products as the result of plastic biodegradation

Plastic	Strain	Degradation time (day)	By-products	Reference	
PET	<i>Streptomyces</i> sp.	18	1) Ethyl benzene 2) o-Xylene 3) 1,2- β Pinene 4) 1,8- Cineole 5) l-Menthone 6) p-Menthan-3-one	7) 2-(4'-Nitro-2'-Thienyl) Pyrimidine 8) Germacrene D 9) Hexamethyl Cyclotrisiloxane 10) 6-Methyl-2-Phenylindole 11) N-Methyl-1-Adamantaneacetamide 12) Cyclohexanital	(Farzi et al. 2019)
PS	<i>Exiguobacterium indicum</i> HHS31	60	1) 2-Pentanone, 4-hydroxy-4-methyl 2) (R)-(-)-2,2-Dimethyl-1,3-dioxolane-4-methanol 3) Oxalic acid, butyl 6-ethyloct-3-yl ester	4) Phenol, 2,4-bis(1,1-dimethylethyl) 5) 1,2-Benzenedicarboxylic acid, butyl 2-methylpropyl ester	(Yang et al. 2015)
PS	<i>Cephalosporium</i> sp.	56	1) Pyridine 2) 1,3,5 Cycloheptatriene 3) Benzene, chloro 4) Methane, tris(methylthio) 5) 1H-Indene, 2,3-dihydro-1,1,3-trimethyl-3-phenyl 6) Benzene, 1,1'-(1,1,2,2-tetramethyl-1,2-ethanediyl) bis	7) 2,4-Diphenyl-4-methyl-2(E)-pentene 8) Pentadecanoic acid, 2,6,10,14-tetramethyl-, methyl ester 9) 13-Tetradecene-11-yn-ol 10) 9-Octadecanoic acid, ethyl ester 11) Octadecanoic acid, ethyl ester	(Chaudhary and Vijayakumar 2020)
PS	<i>Mucor</i> sp.	56	1) n-Hexane 2) Cyclohexane 3) Pyridine 4) 1,3,5 Cycloheptatriene 5) Benzene, chloro 6) Methane, tris(methylthio)	7) 2,4-Diphenyl-4-methyl-1-pentene 8) Benzene, (1,1-dimethyldecyl) 9) Phenol, 2,4-bis(1-methyl-1-phenylethyl) 10) 1H-Indene, 2,3-dihydro-1,1,3-trimethyl-3-phenyl	(Chaudhary and Vijayakumar 2020)
PE	<i>Streptomyces</i> sp.	90	1) 1,4-epoxynaphthalene-1(2h) -methanol,4,5,7-tris(1,1-dimethylethyl) -3 2) 2-t-butyl-5-chloromethyl-3-methyl-4-oxoimidazolidine-1-carboxylic 3) tetrapentacontane	4) Squalene 5) diethyl phthalate 6) benzene, 1,3-bis(1,1-dimethylethyl)	(Abraham et al. 2017)
PE	<i>Aspergillus nomius</i>	90	1) phenol 2) 3,5-bis(1,1-dimethylethyl) 3) 2-t-butyl-5-chloromethyl-3-methyl-4-oxoimidazolidine-1-carboxylic 4) dotriacontane	5) ethyl 14-methyl-hexadecanoate 6) diethyl phthalate 7) benzene, 1,3-bis(1,1-dimethylethyl) 8) dodecane, 1-fluoro 9) difluorophosphoric acid	(Abraham et al. 2017)

Plastic	Strain	Degradation time (day)	By-products	Reference
PE	<i>Aspergillus sydowii</i> strain PNP15/TS	60	1) 7- Methylenebicyclo [3.2.0] hept-3-en-2-one 2) Dibutyl phthalate 3) 1,4-Benzenediol	4) Dodecahydropyrido [1,2-b] isoquinolin-6-one (Sangale et al. 2019)
PE	<i>Aspergillus terreus</i> strain MANF1/WL	60	1) 2 Naphthalene carboxylic acid 2) 2-Cyclohexen 3) Dibutyl phthalate	4) 1,2-Bis (Trimethylsilyl) benzene 5) Hexadecane 6) Hexadecanoic acid (Sangale et al. 2019)
PE	<i>Lysinibacillus fusiformis</i> VASB14/WL	60	1) 1- Trimethylsilylmethanol 2) 1,2,3,4 Tetra methyl benzene 3) Hexadecanoic acid	(Shahnawaz et al. 2016)
PE	<i>Bacillus cereus</i> VASB1/TS	60	1) 1,2,3 Trimethyl benzene 2) 1 Ethyl 3,5-dimethyl benzene 3) 1,4 Di methyl 2 ethyl benzene	4) Dibutyl phthalate 5) Hexadecanoic acid (Shahnawaz et al. 2016)
PE	<i>Pseudomonas aeruginosa</i> PAO1	140	1) Benzene, methyl 2) Tetrachloroethylene 3) Benzene,1,2-dimethyl 4) Octadecane 5) 7,9-Di-tert-butyl-1-oxaspiro (4,5) deca-6,9-dien-2-one 6) Hexadecanoic acid 7) Eicosane 8) Octadecanoic acid	9) Docosane 10) 3-Chloropropionic acid, heptadecyl ester 11) Tricosane 12) Octadecanoic acid, butyl ester 13) 1-Nonadecene 14) Hexacosane 15) Pentacosane 16) 1,2-Benzenedicarboxylic acid, diisooctyl ester (Kyaw et al. 2012)
PE	<i>Aspergillus oryzae</i> strain A5	112	1) 4,4- Dimethyl-2-pentene 2) 4,6- Octadien-3-one,2-methyl	(Muhonja et al. 2018)
PE	<i>Enterobacter</i> sp. D1	31	1) 6-methyl-5-hepten-2-ol 2) Monobenzyl phthalate 3) N-Acetylglutamic acid	4) Ethyldodecanoate (Ren et al. 2019)

Table 5. Continued

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Despite biodegradation is considered the most environmentally friendly way of plastic degradation (Sánchez 2020) with no left toxic and hazardous compounds, one of the important environmental impacts of plastic is the formation of micro (<5 mm) and nano size plastics (0.1 μm < in size). The small plastic particles can be formed as a by-product during biodegradation by the act of microorganisms or can be created after the abandonment of single-used plastics in the environment and exposure with UV, heat, or physical forces. In both cases, the formation of micro and nanoplastics could end up in soil, food, and water systems which may bring several problems for marine and land species.

Napper and Thompson (2016) showed that about 700,000 microplastics particles were entered into our water system and then in the environment after just one cycle of washing machine contained synthetic clothes. While in many developed societies, these particles can be effectively filtered before entering into the water system, still there is a chance of contamination especially in developing countries where such advanced filtration may not even exist.

The toxicity and environmental pollutant of micro and nanoplastics may derive directly from ingestion of the plastic itself, or by contact with other chemicals that associate with microplastics in soil or ocean (Szymańska and Obojowski 2020). While the number of studies for toxicity effect of micro and nano size plastic is still very limited, the adverse effect of micro and nanoplastics were mainly documented in marine and land animals compared to humans except at a high concentration (Yong et al. 2020).

In the study conducted by Ziajahromi et al. (2018) the effect of PE microplastics on survival and growth of *Chironomus tepperi* was analysed and found that a high concentration of PE microplastics affected the growth and shape of *C. tepperi* body. Hwang et al. (2020) studied the potential toxicity of micro and nano PS plastic particles at an approximate dosage of 500ug/mL in human at a molecular level. They found that PS microplastic particles at that dosage had no toxicity effect on humans' cells, however, the particles at nano size (460 nm- 1 μm) were affected red blood cells and led them to hemolysis.

While the mass dispersion of micro and nanoplastics in ocean and soil is problematic, the exact effect of each plastic type at different concentrations is not yet well understood which requires further investigation in both animals and humans.

5. Road to commercialisation

5.1. Technology readiness level

One of the associated challenges in the biodegradation of plastic is the implantation of an effective system to enhance biodegradation efficiencies. Technology readiness levels (TRL) is a well-established system initially developed by NASA to monitor the development of space-based technologies. Since then TRL concept has been widely used in other domains including in the areas of resource recovery. TRL matrix runs from level 1 to 9 and can be categorised into three main scales in terms of the development i) lab scale (level 1-3); ii) pilot scale (level 4-6); and iii) full scale (level 7-9). Fig 6 and Table 6 summarised the TRL levels in each degradation technology and the list of available companies involved in thermal and hydrothermal degradation of plastic was shown. In this instance, the number of applications in TRL 1-3 and 4-6 was determined based on the number of studies in Scopus with the search keywords of technology + plastic + degradation. The number of studies in biodegradation and HTL was determined as biodegradation + plastic; “Hydrothermal liquefaction” + plastic. TRL 7-9 was found based on the number of available companies in the deconstruction of plastic at a commercial (full) scale and the number of registered landfills in the world.

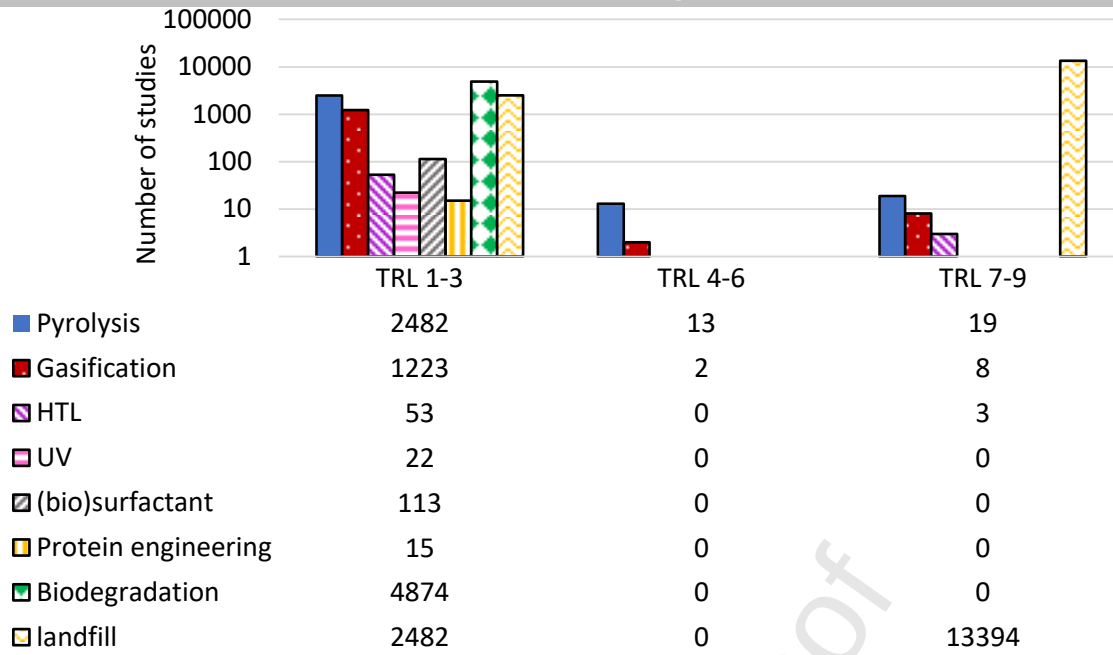


Fig 6. TRL level in different plastic degradation technologies

Table 6. Example of commercial processes in thermal and hydrothermal plastic degradation

Company	Process	Product	Input	Feedstock capacity	Reference
Climax Global Energy	Pyrolysis	Pyrolytic oil	Mixed, post-consumer plastics	-	(Steve Russell 2012)
RES polyflow	Pyrolysis	Diesel, naphtha blend stocks, wax	Mixed, medical automobile, post-consumer plastics	90,000 tonne/year	(Dan Sandoval 2018)
JB1 (plastic2oil)	Pyrolysis	Naphtha, fuel oil	Rigid and Film Plastics (LDPE, HDPE, PP, PS)	1.8 tonne/hour	(Plastic2Oil 2013)
Agilyx	Pyrolysis	Pyrolytic oil	Discrete polymers, mixed	3,600 tonne/year	(Alexander H. Tullo 2019)
Braven Environmental	Pyrolysis	Diesel	Mixed plastics	59,000 tonne/year	(Axel Barrett 2020)
Klean Industries Inc	Pyrolysis	Pyrolytic oil	Mixed plastics	-	(Klean Industries 2020)
Natural State Research	Pyrolysis	Diesel, natural gas	Mixed plastics	-	(Imogene Tyler 2020)
Nexus Fuels	Pyrolysis	Gasoline, kerosene, diesel, heavy oils, waxes	Clean plastics (LDPE, HDPE, PS, PP,)	45 tonne/day	(Nexus Fuels 2019)
Rational Resource	Pyrolysis	Pyrolytic oil	-	-	(ReSource plastic 2020)
Vadxx Energy	Pyrolysis	Diesel, naphtha, waxes	Mixed post-consumer and industrial plastics	54 tonne/day	(Waste today 2020)
Axion Group	Pyrolysis	-	Separated and individual plastic batch	-	(Axion group 2020)
Plastic Energy	Pyrolysis	Diesel, naphtha, pyrolytic oil	Mixed, contaminated, multi-layered plastics	-	(Plastic energy 2020)
Entech - Renewable Energy Solutions	Pyrolysis	Diesel, gasoline, wax, pyrolytic oil	Mixed plastics	-	(Energy expert 2020)
Integrated Green Energy Solutions	Pyrolysis	Diesel, naphtha, marine oil	Waste plastics	31,000 tonne/year	(Sustainable world ports 2018)

Table 6. Continued

Company	Process	Product	Input	Feedstock capacity	Reference
Kanzler Verfahrenstechnik GmbH (KVT)	Pyrolysis	Diesel, gasoline	Sorted and unsorted mixed plastics (PP, PS, LDPE, HDPE)	13,600 tonne/year	(KVT Process Technology 2020)
GEWA Technology	Pyrolysis	High quality fuel	Waste plastics	-	(Environmental expert 2020)
Beston	Pyrolysis	Fuel oil	Waste plastics (LDPE, HDPE, PP, PS)	6-24 tonne/day	(Beston 2020)
Cynar Plc	Pyrolysis	Diesel, pyrolytic oil	Mixed (LDPE, PP, PS)	-	(Process industry Match 2020)
Reaktor ApS	Pyrolysis	Diesel	Contaminated plastics	-	(Denova 2020)
Waste2Tricity	Gasification	High-grade hydrogen to be used as transport fuel	All unrecyclable waste plastics	25 tonne/day	(Thomas Parker 2019)
Showa Denko	Gasification	Obtaining hydrogen for the synthesis of ammonia	Crushed & moulded used plastic	63,000 tonne/year	(Showa Denko 2020)
Texaco	Gasification	Syngas	Mixed plastics	150 tonne/day	(A. Tukker et al. 1999)
Joint cooperation (Shell, Air Liquide, AkzoNobel, Enkema, Nouryon)	Gasification	Syngas and clean methanol for using in the chemical industry	Unrecoverable plastics mixed with waste streams	325,000 tonne/year	(Jim Lane 2019)
ReNew ELP	HTL	Valuable oil, naphtha, heavy waxes	Contaminated, multi-layer flexible and rigid plastics	-	(ReNew ELP)
Neste	HTL	Fuel and new plastics	Waste plastics	Target by 2030 907,000 tonne/day	(Licella 2018)
Licella	HTL	Valuable fuels and chemicals	End-of-life plastics	-	(Licella 2020)

Based on this information the following conclusions can be made:

- 1) Landfilling is a commercially used technology hence a TRL level of 9 can be assigned
- 2) Similarly, Gasification, Pyrolysis, and HTL of plastic is a commercially available technology hence a TRL of 9 can be assigned
- 3) All other methods of plastic degradation are still at a TRL level of 1-3.

As such, from a practical point of view, only the thermal-based plastic degradation processes are currently available as competitive and implementable technologies to the current practice of landfilling.

Although, thermal (ex. pyrolysis and gasification) and hydrothermal approaches (HTL) can currently deconstruct plastic wastes at commercial and pilot scales, the high requirement of energy input, operational expenditures, and high generated toxic gases are the main drawbacks and can make them unattractive options for long-term degrading operation.

5.2 Biodegradation opportunities and limitation

Degradation of plastic with the use of microbial strains (biodegradation) is economically and environmentally beneficial. However, the slow degradation rate due to the lack of proper microbial enzymes is currently the main limitation. Previous studies were mainly focused on the biodegradation of one plastic type (PE), while in the environment a mixture of plastic wastes is exposed for degradation. Enzyme engineering and surface-active agent utilisation have also been reported as potential ways of biodegradation enhancement, our current knowledge and performance in biodegradation are far behind to carry out biodegradation at the pilot and commercial levels.

While Biodegradation is a promising technology in its own right for dealing with plastic wastes and perhaps represents the most advanced and eco-friendly manner to deal with these plastic wastes there are several key limitations and unknowns that must be discussed. The key technical limitation of biodegrading currently is the long periods required for degradation to occur, and the fact that total degradation does not occur. In an attempt to remedy these limitations, there has been a drive toward enhancing natural biodegrading mechanisms and/or through engineered solutions. In both

approaches, enzymatic products specifically are designed to consume certain types of plastics, and genetically modified organisms (GMO's) being developed with the same purpose. However, the use of such GMO's and even modified enzymes means future biodegradation plants and processes systems need to ensure no accidental release of these substances (in particular GMO's) take place. On the other hand, the economic costs of designing and developing such organisms will inevitably increase the cost of developments that is needed to migrate from a low TRL to a medium-high TRL for these pathways. As such, efficiently and economically addressing these shortcomings is an ongoing technical limitation. Another key limitation is the fact that biodegrading of non-degradable thermoplastics mainly occurs on PET due to the availability of discovered enzymes, and it is still limited to the other plastics such as PE, PP and PS.

There have been questions raised in academic circles about the overemphasis of the results achieved in the field of biotechnological plastic degradation, which must also need to be given due consideration. Indeed, the current form of biodegradation technology is unable to deal with the broad spectrum of plastic waste. It is also important to note that, if a specific microorganism is found to have excellent PET degrading capabilities and mediocre for PS, as an example, then commercialisation of such technology would implicitly require the sorting of the plastic waste prior to biodegradation. Alternatively, further development work needs to be done of developing mixed cultures (similar to that of wastewater treatment) that have a combination of microorganisms either in engineering or natural occurring format to degrade different types of plastic at the same time.

5.3 Role of Process Integration

Taking the above-mentioned limitations into account, concepts of process integration from the domain of process systems engineering (PSE) can be applied to somewhat remedy these inherent limitations. One possible approach is to improve the plastic aging process before exposing to biodegradation. This can be done with the help of short hydrothermal treatment at low operating conditions or with the help of UV radiation. As mentioned previously UV has been used as a green pre-treatment technology for aging plastic prior to biodegradation (Roy et al. 2008, Vimala and

Madhew 2010, Monazer et al. 2010). To maximise photo-oxidation by this method, low-wave length (< 300 nm) radiation should be used to obtain maximum penetration in the plastic structure. Irradiation both sides of plastic also is an important factor, which can enhance the again performance. Most of the conducted studies pre-treated plastic samples at a higher wavelength (> 300 nm), and irradiated just one side of the plastic. Irradiation on both side of plastic generates more free-radicals and short-chain compounds in the polymeric structure, which then can increase the chance of microbial attachment and colonisation on the surface of the plastic.

6. Conclusion

Mass production of plastics and plastic wastes undoubtedly is one of the main environmental issues worldwide. Despite the technological advances in the thermal reduction of plastic at a commercial scale, there are still thousands of tonnes remained to be degraded. The unavailability of an efficient and environmentally friendly operational approach has even made the degradation process harder. In this review, the challenges in the common plastic degradation methods were discussed from operational to environmental levels. Plastic biodegradation as an environmentally friendly method has received attention in the last decade, and several plastic degrading strains were identified in this regard. Nevertheless, biodegradation has not been commercialised yet due to its current low efficiencies. Thus, the search for exploring new microbial enzymes and enzyme engineering is still in high demand. In addition to thermal processes and enzyme engineering, the effect of other green methods like UV radiation and surface-active compounds were also studied; however, these approaches alone were not effective enough in improving the degradation efficiency. Therefore, to enhance the degradation rate, the incorporation of processes like hydrothermal and UV prior to biodegradation can be the suitable option for eco and environmentally friendly degradation of plastic in long-term.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Highlights

- Thermal technologies which degrade plastics in full scale are energy intensive.
- Biodegradation is the most environmentally friendly way of plastic degradation.
- The current low biodegradation efficiency limits its potential in commercial scale.
- Enzyme engineering, aging and biosurfactant can enhance biodegradation rate.